Generative Artificial Intelligence

Assignment #2

Question-1: Image Denoising using Denoising Autoencoder

In many real-world imaging applications (medical imaging, surveillance systems, satellite imagery, and mobile photography), images are often corrupted by noise due to sensor limitations or transmission errors. Design and implement a Denoising Autoencoder (DAE) capable of reconstructing clean images from noisy inputs.

The dataset to be used for this task is "The Berkeley Segmentation Dataset and Benchmark"
Dataset Link: https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/

Tasks

1. **Dataset Preparation:** Load the dataset, normalize the images appropriately, and prepare training, validation, and testing splits.
2. **Noise Injection:** Artificially corrupt the training images using two types of noise:
   - Gaussian noise
   - Salt-and-pepper noise
3. **Denoising Autoencoder Model Design:** Design and implement a convolutional denoising autoencoder consisting of encoder and decoder networks. Clearly describe the architecture, bottleneck size, and activation functions used.
4. **Model Training:** Train the denoising autoencoder to reconstruct clean images from noisy inputs using an appropriate reconstruction loss function.
5. **Evaluation and Visualization:** Evaluate the model using suitable reconstruction metrics (e.g., MSE, PSNR, or SSIM). Present visualization results showing:
   - Original clean image
   - Noisy image
   - Reconstructed (denoised) image
6. **Experimental Study:** Analyze how reconstruction performance changes with different noise levels and different bottleneck sizes. Summarize results in tables and figures and discuss observations.

Question-2: Generative Modeling using Variational Autoencoder (VAE)

Generative models aim to learn the underlying data distribution so that new realistic samples can be generated. In this task, students will implement a **Variational Autoencoder (VAE)** to learn latent representations of images and generate new synthetic samples.

The dataset to be used for this task is the Fashion-MNIST dataset.
Dataset Link: Fashion-MNIST Dataset

Tasks

1. **Dataset Preparation** Load and preprocess the Fashion-MNIST dataset, including normalization and preparation of training, validation, and testing sets.
2. **Variational Autoencoder Architecture** Design and implement a VAE consisting of:
   - Encoder network producing latent mean (μ) and variance (σ)
   - Reparameterization trick for latent sampling

- Decoder network reconstructing images from sampled latent vectors

3. **Loss Function Implementation** Implement the combined loss consisting of reconstruction loss and KL-divergence loss.
4. **Model Training** Train the VAE model and monitor both reconstruction loss and KL-divergence during training.
5. **Image Generation and Visualization** Generate new images by sampling random latent vectors from the learned latent space. Also visualize reconstructed images alongside original images.
6. **Experimental Study** Analyze the impact of different latent dimensions on reconstruction quality and generation capability. Present results in tables and figures and discuss observations.

Submission:

- Each student must submit the following three files packaged into a single ZIP file and named as **ROLLNO_NAME.ZIP**:
  - A Jupyter Notebook (.ipynb) or Python script (.py) with the complete implementation.
  - A word/PDF report.
  - A plain text file (.txt) containing all the prompts used for each question.
- Ensure that the code is well-structured with proper comments for each function. Include all necessary dependencies to ensure the code runs without errors.